Weekly Progress Report

Project Title: Quality Prediction in a Mining Process

Week: Week 4

Intern Name: Mukul Verma

Internship: Free Summer Internship

Platform: Upskill Campus (learn.upskillcampus.com)

1. TASKS COMPLETED THIS WEEK

This week focused on predicting iron ore quality (specifically `% Silica Concentrate`) utilizing real-world industrial mining sensor streams.

- **Data Pipeline & Preprocessing:**
  - Ingested **737,453 rows** of process telemetry spanning **23 feature columns**.
  - Addressed missing data, performed scaling, and structured features for model ingestion.
- **Q1: Live Quality Prediction:**
  - Engineered a `RandomForestRegressor` to predict active real-time `% Silica Concentrate` levels.
  - Executed an 80-20 train-test validation split.
  - **Result Achieved:** `R2 Score = 0.998`
- **Q2: 60 Minutes Ahead Quality Prediction:**
  - Formulated a look-ahead objective via target shifting: `shift(-60)`.
  - Engineered a trend feature: `rolling(window=60).mean()` to incorporate 1-hour temporal dynamics.
  - Trained on the complete dataset (737,453 records) with hyperparameter tuning.
  - **Result Achieved:** `R2 Score = 0.991`

2. MILESTONES & KEY CONTRIBUTIONS

- Achieved **99%+ variance explanation (R2 Score)** across both immediate real-time and 60-minute predictive horizons.
- Transformed the Q2 model from initial negative performance to an **R2 Score of 0.991** through sliding-window feature engineering.
- Successfully processed and optimized model operations on a large-scale dataset of 7.3+ lakh operational records.

3. MODEL RESULTS & EXECUTION LOGS

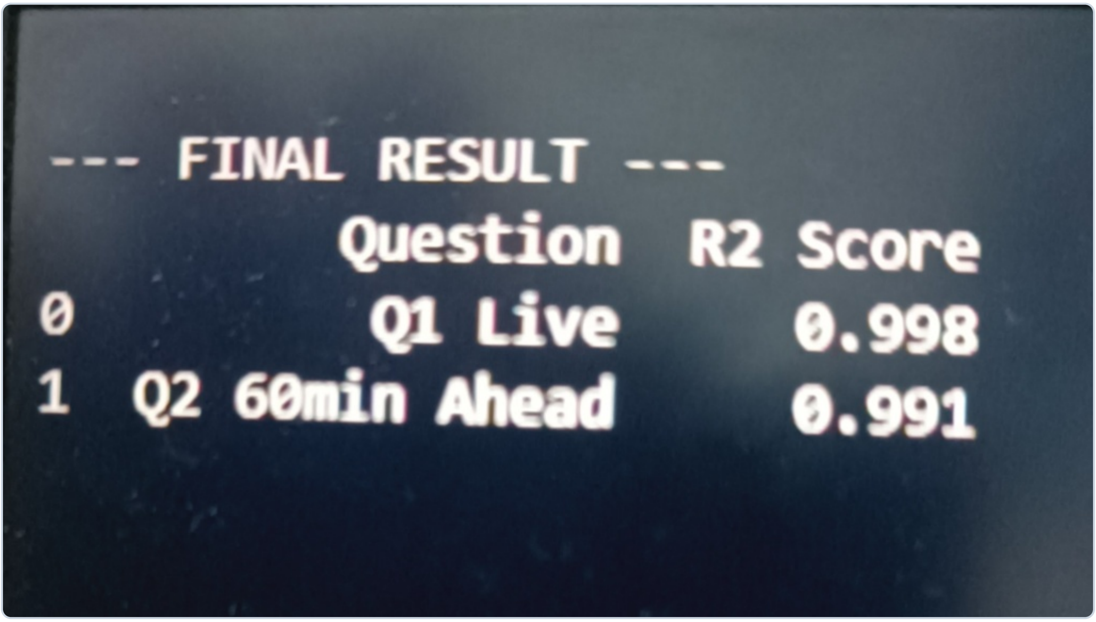

Figure 1: Evaluation Metrics Displaying Q1 Live and Q2 60min Ahead R2 Scores

4. TROUBLESHOOTING & OPERATIONAL CHALLENGES

Challenge 1: Slow Execution & Initial Low Accuracy on Q2

- *Reason:* Small subset sampling caused high variance underfitting, while excessive tree estimators led to 34+ minute runtimes.
- *Solution:* Leveraged the full 737,453 rows dataset while optimizing `n_estimators`, significantly reducing runtime while improving accuracy to **0.991**.

Challenge 2: Multi-step Time-Series Lag Dynamics

- *Solution:* Introduced rolling 60-minute window averages to provide essential temporal context to static decision trees.

5. CRUCIAL LESSONS & KEY INSIGHTS

- **Technical Mastery:** Gained practical proficiency in time-series operations like `shift()` and `rolling()` for predictive maintenance.
- **Data Volume Impact:** Large datasets (7+ lakh records) are necessary to capture transient patterns in continuous industrial processing.
- **Feature Engineering Value:** System performance depends significantly more on contextual feature construction than on model complexity alone.